Does Better Retrieval Mean Better Answers? Benchmarking Multilingual Embeddings for Bangla-English Medical RAG

Trina Chakraborty
Dept. of Computer Science and Engineering
East Delta University
Chattogram, Bangladesh
trinasust5002@gmail.com

Jannatul Naima
Dept. of Computer Science and Engineering
East Delta University
Chattogram, Bangladesh
jannatulnaima.cse.cu@gmail.com

Rim Chowdhury
Dept. of Computer Science and Engineering
Shahjalal University of Science and Technology
Sylhet, Bangladesh
rimchy32@gmail.com

Mst Tuly Khatun
Dept. of Computer Science and Engineering
East Delta University
Chattogram, Bangladesh
tarinn577@gmail.com

***Abstract*—When a retrieval-augmented generation (RAG) system answers a medical question, it can fail two ways: retrieving the wrong document, or retrieving the right one but generating an answer it doesn't actually support. Whether multilingual retrievers handle this equally across languages is untested for a low-resource language like Bangla, especially in medicine: prior work reports retrieval or generation quality alone, rarely both together under matched conditions. We build a parallel English-Bangla dataset on PubMedQA: 200 biomedical questions with gold passages, each translated into Bangla via NLLB-200, giving 400 queries that share one 200-passage corpus and identical gold labels, so language is the only variable that changes. Our benchmarking framework scores every query on two connected stages: three multilingual retrievers (bge-m3, multilingual-e5-base, LaBSE) are evaluated by Recall@k, MRR, and nDCG, and their retrieved passages are passed to one shared generator whose answer is scored for faithfulness by a multilingual NLI model. Retrieval quality drops from English to Bangla for every retriever, but unevenly and with statistically significant gaps between models (4 points of Recall@1 for bge-m3 versus 21 for multilingual-e5-base), and this gap carries through to generation, with Bangla answers 14–27 points less faithful. To isolate retrieval error from a generator- or metric-side effect, we add an oracle condition given the guaranteed-correct passage: it still shows a 16.5-point gap, comparable to bge-m3's real 14.4-point gap, indicating retrieval error alone does not account for the drop. bge-m3 is the most language-robust retriever tested and the most reliable choice across both languages.**

***Index Terms*—retrieval-augmented generation, dense retrieval, multilingual natural language processing, low-resource languages, Bangla, medical question answering, faithfulness evaluation**

I. Introduction

Retrieving the correct supporting passage from a knowledge source is a fundamental task in information retrieval, providing the evidence needed by downstream applications such as open-domain QA, fact verification, and RAG. Modern multilingual dense retrievers are highly effective at finding relevant passages across languages, but are mostly evaluated on English or other high-resource benchmarks, leaving it unclear whether they provide the same retrieval quality for low-resource languages such as Bangla, spoken by more than 230 million people yet underrepresented in language resources. This matters most in medicine: a retriever that performs poorly for a language can fail silently, since the generator still produces a fluent, convincing answer that may not be supported by the correct medical evidence.

Existing RAG evaluation practice compounds this risk: retriever comparisons are usually reported using retrieval metrics alone (Recall, MRR), without connecting them to the faithfulness of the answers a user actually reads – an empirical question not, to our knowledge, examined for Bangla medical RAG specifically.

A short example illustrates the stakes: a biomedical question is machine translated into Bangla while its correct supporting abstract stays fixed, and two retrievers must retrieve that abstract for both versions. bge-m3 ranks the correct passage first in both languages, but multilingual-e5-base, also first in English, drops it to rank 3 in Bangla, even though the question's meaning is unchanged – motivating a systematic, language-stratified evaluation.

Motivated by this observation, we present a controlled, language stratified benchmark that jointly evaluates retrieval quality and downstream answer faithfulness for Bangla and English medical question answering. Given only a question (never the supporting document), each retriever must find the correct passage from a fixed 200-passage corpus; the retrieved passages are then handed to a language model that generates an answer. We measure both steps separately, for the same questions asked in English and Bangla, so any

difference we observe is caused by question language alone. Our contributions are as follows.

- We evaluate three multilingual dense retrievers: bge-m3, multilingual-e5-base, and LaBSE, on the same 200 biomedical questions in both English and Bangla. Their retrieval performance is compared using Recall@1, Recall@5, Mean Reciprocal Rank (MRR), and Normalized Discounted Cumulative Gain (nDCG) [1].
- We generate an answer from each retriever's output using the same language model and prompt, and score how well that answer is supported by the retrieved passages using an NLI entailment metric, so we can tell whether a retrieval gap actually reaches the final answer or stays contained at the retrieval step.
- We show empirically that it does not stay contained: for all three retrievers, the drop in answer faithfulness from English to Bangla (14–27 points) is larger than the corresponding drop in retrieval Recall@1, and that the retriever with the best retrieval score (bge-m3) is also the one with the smallest English-to-Bangla drop at both steps.
- We isolate how much of this drop is caused by retrieval error itself, versus the generator or faithfulness metric, with an oracle baseline (guaranteed-correct passage) and a random-passage lower bound; we further situate the dense retrievers against a sparse BM25 baseline and a translate-then-retrieve pipeline, and confirm which pairwise differences are statistically reliable via paired bootstrap tests.
- Based on these results, we recommend bge-m3 over multilingual-e5-base and LaBSE specifically for Bangla-facing medical RAG, since aggregate (English-dominated) benchmarks would not have revealed this difference.

II. Related Work

A. Retrieval-Augmented Generation

RAG improves the factual reliability of LLMs by incorporating external knowledge during generation. Lewis *et al.* [2] proposed the RAG framework, combining DPR with a sequence-to-sequence generator for knowledge-intensive QA. Gao *et al.* [3] surveyed RAG systems and emphasized that effective retrieval underlies reliable grounded generation.

B. Dense and Multilingual Retrieval

DPR [4] established dense retrieval via dual-encoder representations; ANCE [5] added dynamic hard-negative sampling, and ColBERT [6] introduced efficient late-interaction matching. For multilingual retrieval, LaBSE [7], Multilingual-E5 [8], and BGE-M3 [9] advanced cross-lingual representation learning. MTEB [10] and MMTEB [11] provide standard benchmarks, evaluated almost entirely on high-resource languages, leaving Bangla retrieval quality largely untested.

C. Medical QA and Healthcare RAG

PubMedQA [12] and BioASQ 2025 [13] standardize biomedical QA and retrieval evaluation. Xiong *et al.* [14] showed retriever design significantly affects medical RAG accuracy, and Yang *et al.* [15] emphasized reliable retrieval for trustworthy healthcare AI, but existing studies focus mainly on English.

D. Faithfulness Evaluation in RAG

Reliable retrieval does not guarantee faithful responses. RAGAS [16] introduced automated faithfulness and context metrics; Yu *et al.* [17] surveyed evidence-grounded evaluation, and MEMERAG [18] extended it to multilingual RAG, but these too are built almost entirely on high-resource languages, leaving the link between embedding choice and faithfulness underexplored for low-resource medical settings.

E. Bangla and Low-Resource Medical NLP

BanglaBERT [19] advanced Bangla language understanding via large-scale pretraining, and IndicBART [20] introduced a multilingual seq2seq model for 11 Indic languages, showing effective low-resource transfer. Their integration with multilingual retrieval for medical RAG remains limited; existing work rarely jointly evaluates retrieval quality and answer faithfulness for English-Bangla medical RAG, the gap this paper addresses.

III. Proposed Evaluation Framework

Our evaluation framework addresses language stratified RAG assessment through three sequential stages: (1) parallel corpus and query construction across English and Bangla, (2) multilingual dense retrieval and evaluation, and (3) grounded answer generation with natural language inference based faithfulness scoring. Fig. 1 summarizes this pipeline; each stage is described in the following subsections.

How the two stages connect. Retrieval and generation use different inputs. In *retrieval*, the system is given only a question – never the document that answers it – and ranks all 200 corpus passages by likelihood of containing the answer; we check whether the gold passage lands at or near the top (Section III-B). In *generation*, we take the retriever's top three passages, whether or not they include the correct one, and give them to a language model with the question, instructing it to answer using only that text, in the question's own language (Section III-A); we then check whether that answer is actually supported by what it was given, via the faithfulness metric (Section III-D). In short: retrieval is scored by whether the right passage was found, generation by whether the answer stays faithful to whatever was found.

A. Parallel Corpus and Query Construction

We build our evaluation on PubMedQA [12], a biomedical question answering dataset in which each question is paired with the abstract context from which it was derived. We use the expert annotated labeled split and sample 200 questions with a fixed random seed. For each question, the associated abstract

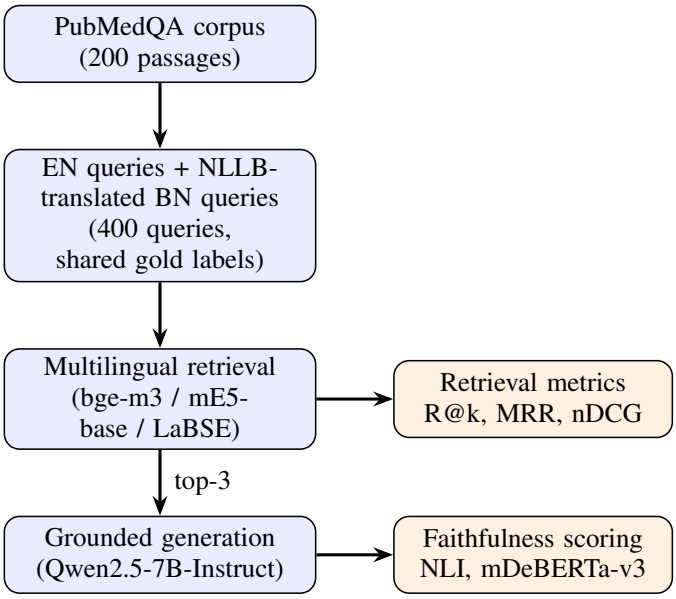


Fig. 1. Overview of the proposed evaluation pipeline. The two scored stages are visually separated: retrieval metrics branch off the retrieval step, faithfulness scoring branches off the generation step.

forms one passage in the retrieval corpus, giving a corpus of 200 passages in which every query has exactly one known gold passage, yielding retrieval relevance labels automatically without manual annotation.

To obtain a parallel Bangla query set, we machine translate the 200 English questions into Bangla using NLLB-200 distilled with 600 million parameters [21]. Only the *questions* are translated; the 200-passage corpus stays English in both conditions, since translating the abstracts would introduce a second, confounding source of noise. Each Bangla question inherits its gold label from its English original, so both language conditions are evaluated against the same corpus and gold labels; only the question language differs. The final evaluation set contains 400 queries (200 English, 200 Bangla), each with a known correct passage. The Bangla queries are machine translated rather than native, which we treat as a limitation in Section IV-G.

B. Multilingual Retrieval

We evaluate three multilingual embedding models: bge-m3, multilingual-e5-base, and LaBSE. Passages and queries are encoded with L2 normalized embeddings and ranked by cosine similarity, prepending E5's required instruction prefixes. To control computational cost we cap the maximum sequence length at 512 tokens for all models. We report Recall@1/3/5, MRR, and nDCG@1/3/5 overall (Table I) against the single gold passage per query, and Recall@1 and MRR stratified by query language (Table II).

The three models differ substantially in architecture: bge-m3 (560M params, 1024-dim, XLM-R-large) natively supports 8192 tokens; multilingual-e5-base (278M, 768-dim, XLM-R-base) supports 512; LaBSE (471M, 768-dim, BERT dual encoder) supports only 256, shorter than our 512-token cap – a confound we return to in Section IV-G.

C. Grounded Generation

For each query we take the retriever's top three ranked passages (regardless of whether the correct one is among them) and give them, with the original question, to an instruction-tuned generator, Qwen2.5-7B-Instruct [22], running in half precision and told to answer using only the given passages, not outside knowledge. The answer language always matches the question language, even though the passages are always English (Section III-A). We use a single fixed prompt for every query: *"You are a careful medical assistant. Answer the question using ONLY the context below. If the context is insufficient, say you do not know. Answer in the same language as the question. Context:* {*context*} *Question:* {*question*} *Answer:"*, with greedy decoding (`do_sample=False`) and up to 256 new tokens, identical across retrievers.

D. Faithfulness Scoring

This subsection defines the faithfulness metric itself, as part of the method; the actual faithfulness scores obtained for each retriever are reported later, in Section IV.

We measure answer faithfulness with a multilingual natural language inference cross-encoder based on mDeBERTa-v3-base fine tuned for cross-lingual natural language inference [23]. We split each generated answer into sentences, correctly handling the Bangla sentence delimiter known as the danda in addition to Latin sentence punctuation. For each answer sentence we compute its maximum entailment probability against the retrieved context passages, and define the answer's faithfulness as the mean of these per sentence maxima:

$$\mathrm{Faith}(a \mid C) = \frac{1}{|S(a)|} \sum_{s \in S(a)} \max_{c \in C} P_{\mathrm{ent}}(c \Rightarrow s), \qquad (1)$$

where $S(a)$ is the set of sentences in answer a, C is the set of retrieved contexts, and P_{ent} is the entailment probability from the natural language inference model. Higher values indicate answers more fully supported by the retrieved evidence.

IV. Experiments and Evaluation

A. Overall Retrieval Performance

Q: How do the three multilingual retrievers compare in absolute biomedical retrieval quality?

Table I reports overall retrieval performance across all 400 queries. The three models separate cleanly: bge-m3 is strongest across every metric (0.975 R@1, 0.982 MRR), multilingual-e5-base is intermediate (0.888 R@1, 0.909 MRR), and LaBSE trails substantially (0.622 R@1, 0.710 MRR). This ordering recurs throughout the remaining analyses and is visualized in Fig. 2(a).

TABLE I
OVERALL RETRIEVAL PERFORMANCE (400 QUERIES)

Model	R@1	R@3	R@5	MRR	nDCG@1	nDCG@3	nDCG@5
bge-m3	**0.975**	**0.988**	**0.990**	**0.982**	**0.975**	**0.983**	**0.984**
multilingual-e5-base	0.888	0.930	0.938	0.909	0.888	0.913	0.916
LaBSE	0.622	0.798	0.822	0.710	0.622	0.728	0.739

B. The English-to-Bangla Retrieval Gap

Q: Does retrieval quality transfer equally from English to Bangla, or does it depend on which retriever is used?

Table II decomposes retrieval performance by query language. Every model performs worse on Bangla than on English, but the magnitude of the drop differs sharply between models – the central finding of this analysis. bge-m3 is remarkably robust, dropping only 4 points in R@1 (0.995→0.955). multilingual-e5-base, despite near-perfect English performance (0.995), collapses on Bangla to 0.780, a 21.5-point gap. LaBSE is weak in both languages, dropping from 0.700 to 0.545. The overall ranking in Table I thus conceals a second, language-robustness axis on which multilingual-e5-base in particular performs far worse than its strong English numbers suggest.

TABLE II
RETRIEVAL BY QUERY LANGUAGE (N = 200 EACH)

	English		Bangla	
Model	R@1	MRR	R@1	MRR
bge-m3	0.995	0.995	**0.955**	**0.969**
multilingual-e5-base	0.995	0.997	0.780	0.820
LaBSE	0.700	0.793	0.545	0.627

C. Does Retrieval Quality Predict Downstream Faithfulness?

Q: Does the retrieval ordering observed above carry through to the faithfulness of the generated answers, and does the English to Bangla gap compound or stay contained at the retrieval stage?

Table III reports faithfulness overall and by language. Two findings stand out. First, overall faithfulness tracks retrieval quality in rank order: bge-m3 is most faithful at 0.598, followed by multilingual-e5-base at 0.585 and LaBSE at 0.554, the same ordering as Table I – though Section IV-F's significance tests show only the differences against LaBSE are statistically reliable; bge-m3 vs. multilingual-e5-base is not significant. Retrieval metrics, inexpensive to compute, therefore carry real but limited predictive signal here. Second, faithfulness degrades from English to Bangla for every model, and more sharply than the corresponding retrieval degradation: 14 points for bge-m3, 18 for multilingual-e5-base, and 27 for LaBSE. The gap therefore does not stay contained at retrieval: it compounds through generation, so a retriever's language weakness is reflected, and usually amplified, in the faithfulness of the final answer a user reads.

Third, this qualifies rather than overturns the first finding: LaBSE, the weakest retriever overall, has the *highest* English faithfulness (0.688), ahead of bge-m3 (0.670). LaBSE's native 256-token context (Section III-B) is shorter than the 512-token cap applied to all models, so it likely retrieves shorter, more topically narrow passages, which can inflate NLI entailment regardless of correctness – an anomaly that motivates the oracle analysis below.

TABLE III
ANSWER FAITHFULNESS (MEAN NLI ENTAILMENT)

Model	Overall	English	Bangla
bge-m3	**0.598**	0.670	**0.526**
multilingual-e5-base	0.585	0.676	0.494
LaBSE	0.554	**0.688**	0.421

D. Qualitative Analysis

Q: What do individual failure cases look like, and what do they suggest about the source of the English to Bangla gap?

To complement the aggregate metrics above, we inspected individual Bangla queries where a retriever's rank for the gold passage worsened relative to its English counterpart. Two recurring patterns emerged. The first is genuine cross-lingual retrieval difficulty: for domain specific biomedical terminology with no common Bangla clinical equivalent, cosine similarity between the Bangla query embedding and the correct passage is measurably weaker than the English-to-English similarity, consistent with Table II's gap being a real alignment weakness rather than ranking noise.

The second is translation noise: manual inspection surfaced clinical terminology errors, including "aortic" mistranslated to a term meaning "Arctic", and "abdomen" rendered with an unrelated compound term; here a rank drop partly reflects a corrupted query rather than a genuine retriever limitation. We treat the two patterns' relative contribution as an open question, discussed in Section IV-G.

E. Isolating the Source of the Gap: An Oracle Baseline

The results above show the gap compounds through generation, but not how much is caused by retrieval error itself versus a generator- or metric-side effect that would persist even with perfect retrieval. To isolate this, we add an oracle condition bypassing retrieval entirely (generator given only the single guaranteed-correct gold passage) and a random condition (three non-gold passages) as a lower bound.

The oracle uses one passage rather than three, and since faithfulness takes a max over the context set (Eq. 1), this mechanically disadvantages its absolute score, which is why oracle English faithfulness (0.609) falls below all three real retrievers despite always retrieving the correct evidence; both language conditions share this handicap symmetrically, so the *gap* comparison remains valid. The oracle gap (16.5 points) is close to bge-m3's real gap (14.4 points) and well above zero, indicating a substantial share of the gap persists even

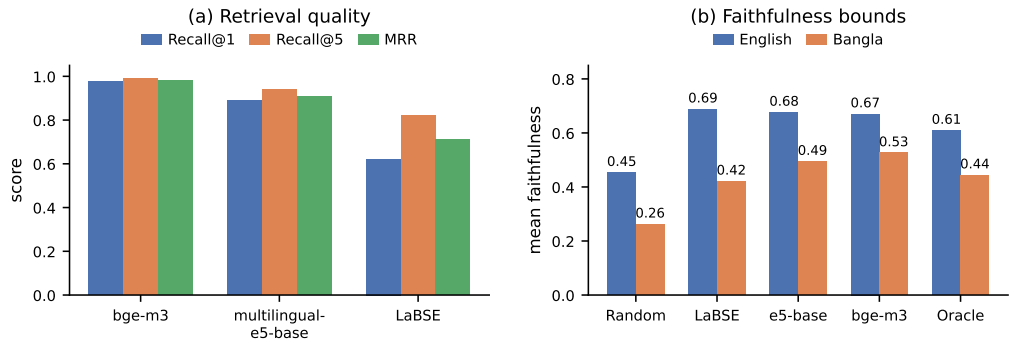


Fig. 2. (a) Overall retrieval quality per model. (b) Faithfulness bounds: random (lower bound), the three real retrievers, and the oracle (gold passage, upper bound), by query language. The aggregate faithfulness ordering (Table III) matches retrieval quality, though this reverses in the English-only subset (Sec. IV-C); the oracle gap (16.5 pts) is comparable to bge-m3's real gap (14.4 pts, Sec. IV-E).

TABLE IV
FAITHFULNESS BOUNDS: RANDOM, REAL RETRIEVERS, AND ORACLE

Condition	English	Bangla (gap)
Random (3 passages)	0.453	0.262 (19.1)
LaBSE	0.688	0.421 (26.7)
multilingual-e5-base	0.676	0.494 (18.2)
bge-m3	0.670	0.526 (14.4)
Oracle (1 gold passage)	0.609	0.444 (16.5)

with perfect, language-invariant retrieval – most plausibly from the generator or NLI scorer behaving differently by answer language.

F. Additional Baselines and Statistical Validation

To situate the dense retrievers against non-neural and pipeline alternatives, and to test whether the observed differences are statistically reliable, we report three additional analyses: a sparse lexical baseline (BM25); a translate-then-retrieve baseline that back-translates Bangla queries to English with NLLB and retrieves monolingually with bge-m3; and pairwise significance tests among retrievers.

TABLE V
SPARSE AND TRANSLATION BASELINES (RECALL@1 / @5 / MRR)

Condition	R@1	R@5	MRR
BM25 (English)	0.960	0.980	0.969
BM25 (Bangla)	0.095	0.110	0.105
bge-m3, direct (Bangla)	0.955	0.985	0.969
NLLB back-translate + bge-m3 (Bangla)	0.970	0.990	0.978

BM25 performs almost as well as bge-m3 on English (0.960 vs. 0.995 R@1) but collapses on Bangla (0.095), since lexical overlap needs shared vocabulary, confirming the gap is not solely a dense-embedding artifact. Translating Bangla queries to English before retrieval (0.970 R@1) slightly outperforms bge-m3's direct cross-lingual retrieval (0.955), suggesting a small residual alignment cost, though the pipeline trades this for translation latency and error propagation. Paired bootstrap tests (10,000 resamples) give 95% CIs and two-sided p-values. All three Recall@1 differences are significant with CIs excluding zero: bge-m3 $-$ e5-base $= 0.088$ [0.060, 0.118], bge-m3 $-$ LaBSE $= 0.353$ [0.305, 0.400], e5-base $-$ LaBSE $= 0.265$ [0.215, 0.313] (all $p < 0.001$). For faithfulness, bge-m3 $-$ LaBSE $= 0.044$ [0.015, 0.073] ($p = 0.002$) and e5-base $-$ LaBSE $= 0.031$ [0.002, 0.060] ($p = 0.035$) are significant, but bge-m3 $-$ e5-base $= 0.013$ [-0.014, 0.040] ($p = 0.34$) is not. A truncation audit against the shared 512-token embedding cap found only 4.0%, 4.0%, and 2.0% of passages exceeded it for bge-m3, e5-base, and LaBSE respectively (mean 327.6, 327.6, and 297.8 tokens) – too small a share to explain the retrieval gaps. This understates LaBSE's real truncation, however: its own native limit is 256 tokens (Section III-B), and its passages average 297.8 tokens, already above that, so a substantial share are truncated for LaBSE specifically in a way the 512-token number misses – consistent with Section IV-C's context-length explanation.

G. Threats to Validity

Several factors should be considered when interpreting our results.

Machine-translated queries. Our Bangla queries come from NLLB-200 machine translation, not native speakers. As shown in Section IV-D, manual inspection surfaces occasional clinical terminology errors, so the Bangla condition partly measures translation-artifact robustness in addition to genuine cross-lingual difficulty; our reported gaps should be read as an upper bound relative to clean, native Bangla queries. Native or human-verified queries are needed to isolate the two effects.

Faithfulness is a proxy, and its cross-lingual calibration is unverified. We operationalize faithfulness as NLI entailment of answer sentences by retrieved contexts – reference-free and scalable, but NLI models can miss valid paraphrastic support or reward lexical overlap, and their calibration may differ across languages, itself potentially contributing to the gap independent of retrieval quality; the oracle analysis (Section IV-E) is consistent with a nontrivial metric- or generator-side share. We have not yet validated the metric against human

judgments in either language, and log this as future work ($\sim$50 annotated samples per language) rather than assume comparable calibration. Our scores should be treated as relative comparisons under one fixed metric, not an absolute measure of correctness.

Scope. All experiments use one biomedical QA dataset, a 200-passage corpus with a single gold passage per query, and one generator; results may not transfer to other domains, larger corpora, multi-passage relevance, or other generators, and with 200 questions per language, small model differences warrant caution.

V. Conclusion and Future Work

We presented a language stratified evaluation framework that jointly measures retrieval quality and downstream answer faithfulness for Bangla and English biomedical RAG, and applied it to benchmark three multilingual embedding models. Retrieval quality predicts downstream faithfulness at the level of relative model ranking; every retriever degrades from English to Bangla; and this degradation compounds through generation rather than staying contained at retrieval. An oracle baseline with guaranteed-correct evidence still shows a 16.5-point gap, comparable to bge-m3's real 14.4-point gap, indicating retrieval error alone does not explain the full drop. bge-m3 is both the strongest and most language-robust retriever tested, with retrieval advantages significant at every comparison (Section IV-F) – though its faithfulness advantage is significant only against LaBSE, so the recommendation rests on retrieval robustness first. multilingual-e5-base illustrates the risk of evaluating on English alone: near-perfect English retrieval conceals a substantial Bangla weakness visible only under language-stratified evaluation.

More broadly, our findings caution against treating multilingual retrievers as language agnostic in safety-sensitive, low-resource deployments. Future work will replace machine-translated queries with native, human-verified Bangla, run a human evaluation to validate the NLI metric's cross-lingual calibration (Section IV-G), evaluate answer correctness against reference answers, and extend this framework to other domains and larger corpora. We will release our code and data upon publication to support reproducibility.[1]

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[1]HF model IDs: retrievers BAAI/bge-m3, intfloat/multilingual-e5-base, sentence-transformers/LaBSE; generator Qwen/Qwen2.5-7B-Instruct (fp16, greedy, do_sample=False, max 256 new tokens); translation facebook/nllb-200-distilled-600M (greedy, max 256 new tokens); faithfulness scorer MoritzLaurer/mDeBERTa-v3-base-xnli-multilingual-nli-2mil7 (fp16). Seed 42 throughout. Hardware: single NVIDIA A100-SXM4-40GB GPU.

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